

Q1.2:

1. Report the solution path and number of node expansions for DFS and BFS.
 - a. DFS
 - i. [(3, 3, 0, 0, 'L'), (2, 2, 1, 1, 'R'), (3, 2, 0, 1, 'L'), (3, 0, 0, 3, 'R'), (3, 1, 0, 2, 'L'), (1, 1, 2, 2, 'R'), (2, 2, 1, 1, 'L'), (0, 2, 3, 1, 'R'), (0, 3, 3, 0, 'L'), (0, 1, 3, 2, 'R'), (0, 2, 3, 1, 'L'), (0, 0, 3, 3, 'R')]
 - ii. 12 Node expansions
 - b. BFS
 - i. [(3, 3, 0, 0, 'L'), (3, 1, 0, 2, 'R'), (3, 2, 0, 1, 'L'), (3, 0, 0, 3, 'R'), (3, 1, 0, 2, 'L'), (1, 1, 2, 2, 'R'), (2, 2, 1, 1, 'L'), (0, 2, 3, 1, 'R'), (0, 3, 3, 0, 'L'), (0, 1, 3, 2, 'R'), (1, 1, 2, 2, 'L'), (0, 0, 3, 3, 'R')]
 - ii. 15 Node expansions
2. Write a short free-form explanation (1–2 paragraphs) comparing DFS and BFS in the context of this problem:
 - a. DFS and BFS are similar algorithms with the intent to traverse a tree either by a stack (LIFO) or a queue (FIFO). Although both options have the same total cost to reach the optimal solution (moving all missionaries and cannibals to the right side of the riverbank), BFS has a higher number of node expansions. In most cases, BFS would return an optimal solution for every iteration compared to DFS due to the nature of the algorithms. We see the DFS expansion count is lower than that of the BFS, meaning DFS used less memory and still reached an optimal solution. This can be accredited to the constraints given in the original river crossing problem that forces the cannibals to never outnumber the missionaries and the boat to carry two people max.

Q2.2:

1. Run UCS with Cost Model A and Cost Model B,
 - a. UCS model A for load discrimination
 - i. [(3, 3, 0, 0, 'L'), (3, 1, 0, 2, 'R'), (3, 2, 0, 1, 'L'), (3, 0, 0, 3, 'R'), (3, 1, 0, 2, 'L'), (1, 1, 2, 2, 'R'), (2, 2, 1, 1, 'L'), (0, 2, 3, 1, 'R'), (0, 3, 3, 0, 'L'), (0, 1, 3, 2, 'R'), (0, 2, 3, 1, 'L'), (0, 0, 3, 3, 'R')]
 - ii. Total cost of 23
 - iii. 15 Node expansions
 - b. UCS model B for travel discrimination
 - i. [(3, 3, 0, 0, 'L'), (2, 2, 1, 1, 'R'), (3, 2, 0, 1, 'L'), (3, 0, 0, 3, 'R'), (3, 1, 0, 2, 'L'), (1, 1, 2, 2, 'R'), (2, 2, 1, 1, 'L'), (0, 2, 3, 1, 'R'), (0, 3, 3, 0, 'L'), (0, 1, 3, 2, 'R'), (0, 2, 3, 1, 'L'), (0, 0, 3, 3, 'R')]
 - ii. Total cost of 17
 - iii. 15 Node expansions
2. Compare with BFS:
 - a. UCS follows a differing sequence of actions en route to the goal state compared to BFS. The 2nd from first and 2nd from last states for UCS are different when comparing model B to BFS, while only the 2nd to last state is different when comparing BFS to model A. Both models share the same amount of node expansions as BFS
3. Explain why UCS is necessary with non-uniform costs, and discuss the trade-off between least cost and fewest steps
 - a. UCS requires (strictly necessary) the cost for each available action to be non-uniform as a way to differentiate itself from BFS. The idea that BFS finds the shortest path to be the best path gets expanded on with the UCS algorithm since not all scenarios with multiple actions and states can be represented in an all-costs-are-equal system without losing sight of the

overall problem. UCS targets a range of outcomes that minimize the step and cost variables instead of the depth of the tree, ensuring an optimally “cheap” solution.

Q3.2:

1. Admissibility proofs

a. H1 (Passenger Weight Remaining)

- i. Given: $h_1(s) = 2M_{left} + 1C_{left}$
- ii. The rules of the game state that everyone must move from the left side of the riverbank to the right side at least once during the simulation. H1 gives a cost function of 2 units vs 1 unit to move a missionary vs. a cannibal, but the function fails to consider the travel cost of the boat (as seen in Model B), which means this heuristic will always underestimate the true remaining cost of the scenario, given the heuristic H1. Therefore, we can conclude that H1 is admissible.

b. H2 (Trip-Packing Lower Bound)

- i. Given: $h_2(s) = \lceil \frac{2M_{left} + 1C_{left}}{3} \rceil$
- ii. We have already proven that H1 is admissible, and can observe that it is contained within H2. Since the only change is to implement a ceiling for the function, H2 implicitly must have a smaller cost estimation than H1. Since H1 is admissible and $h_2(s) \leq h_1(s)$, we can conclude that H2 is also underestimating the true cost, and is an admissible heuristic.

2. Comparison table

Heuristic	Solution Path	Total Cost	Node Expansions	Same cost as UCS?
H1	See Output	23	14	Yes
H2	See Output	23	15	Yes
H3				

3. Designing h3

Q3.3

Extra credit proofs

4. H1 (let $H = 2m + 1c$)

- $h_1(s') = h_1(s) - H$ for Left to Right boat travel
- $h_1(s) \leq H + (h_1(s) - H)$
- $h_1(s) \leq h_1(s)$, therefore H1 is consistent moving left to right
- $h_1(s') = h_1(s) + H$ for Right to Left boat travel
- $h_1(s) \leq h + h_1(s) + H$
- $0 \leq 2H$
- H is an always positive quantity by definition. Therefore H1 is fully consistent

5. H2